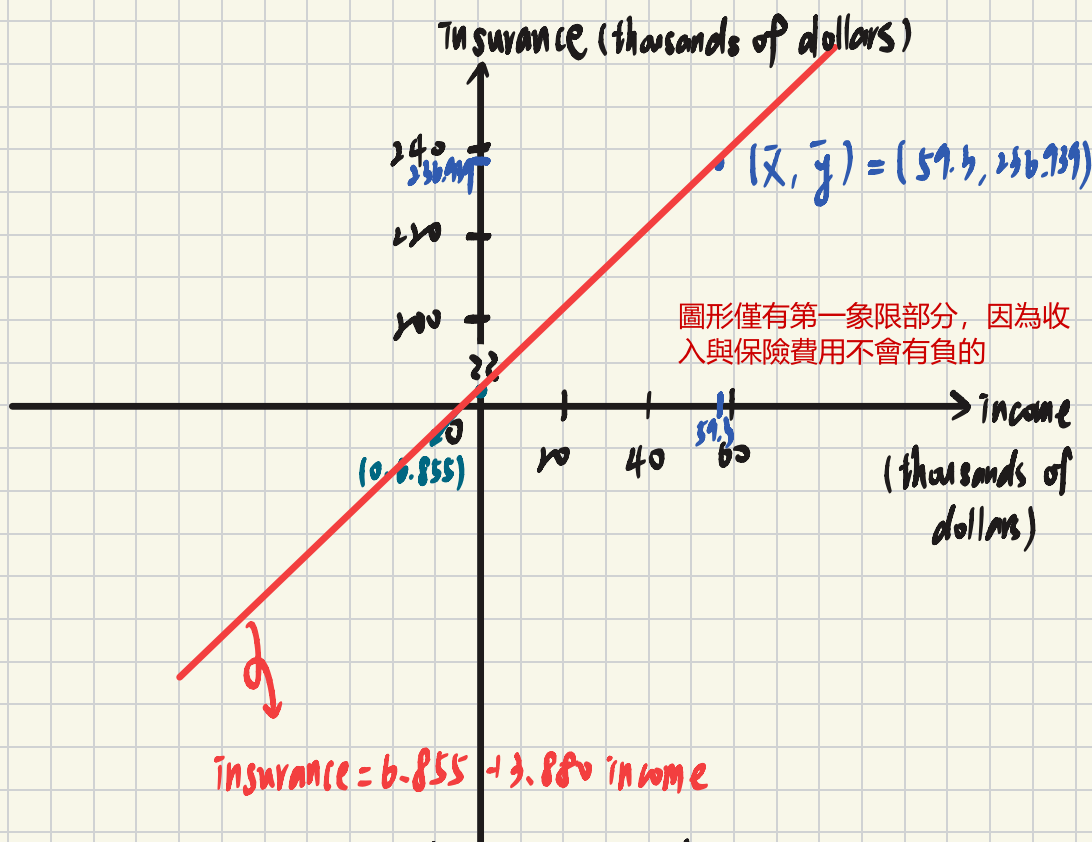


3.18 $\widehat{INSURANCE} = 6.855 + 3.880 INCOME, N=10$
 (se) (7.383) (0.112)

a. sample mean of income = 59.3. 代入 fitted regression line $\Rightarrow$ sample mean of the amount of insurance held
 $= 6.855 + 3.880 \times 59.3 = 236.939$



The slope of the fitted regression line is 3.880 and the intercept is 6.855

b. When each additional \$1,000 of household income changes, the average amount of insurance held will change. according to the fitted regression line, $(1 = 3.880 \times 1000)$ is \$3,880, which is the slope of the line.

The point estimate is 3.880, and the 95% interval estimate is $3.880 \pm t_{\alpha/2} \times se(b_1)$, where $\alpha = 0.05$, degree of freedom = $20 - 2 = 18$

查表得, $df=18$, 右尾面積 = 0.025 之 t 值

$$\therefore 95\% CI = 3.880 \pm \underline{1.101} \times 0.112 = 3.880 \pm 0.235312$$
$$= [3.644688, 4.115312]$$

95% CI 的意思為, 當 household income 每增加 \$1,000, 會有 95% 的區間包含到母體參數, 也就是 insurance 真正會改變的數額

c. When a household with \$100,000 income, the point estimate of the insurance is

$$6.855 + 3.880 \times 100 = 6.855 + 388 = 394.855$$

$$\begin{aligned} \text{the variance of insurance} &= \text{Var}(b_1) + 100^2 \text{Var}(b_2) \\ &+ 2 \times 1 \times 100 \text{Cov}(b_1, b_2) = 7.385^2 + 100^2 \times 0.112^2 + 2 \times \\ &100 \times (-0.746) \doteq 30.7487 \end{aligned}$$

$$\Rightarrow \text{se}(\text{insurance}) = \sqrt{\text{Var}(\text{insurance})} \doteq 5.5452$$

As a result, the 99% CI ($\alpha=0.01$) is

$df=18$, 右尾面積 = 0.005 $\rightarrow$ t-value

$$394.855 \pm \underline{2.878} \times 5.5452 = [378.8936, 410.8164]$$

$$d. \text{INSURANCE} = \beta_1 + \beta_2 \text{INCOME} + \varepsilon$$

$$\alpha = 0.05$$

假設檢定 β_2 是否為 5

$$\Rightarrow \begin{aligned} H_0: \beta_2 &= 5 \quad (\text{雙尾檢定}) \\ H_a: \beta_2 &\neq 5 \end{aligned}$$

$$\text{檢定統計量 } t \text{ 值} = \frac{4.880 - 5}{0.112} = -10$$

$$\text{又 critical value } t_c = t_{0.025, df=18} = \pm 2.101$$

$$\therefore \text{統計量 } t = -10 < -t_c = -2.101$$

$\therefore$ 拒絕 H_0 , 推測 H_a 為真

因此, 資料並不支持該管理層的說法

e. $H_0: \beta_2 = 1$ (one-tailed test), $\alpha = 0.01$
 $H_a: \beta_2 > 1$

檢定統計量 $t = \frac{1.880 - 1}{0.112} = 15.7143$

又 critical value $t_c = t_{0.005, df=18} = 2.552$

$\therefore$ 統計量 $t = 15.7143 > t_c = 2.552$

$\therefore$ 拒絕 H_0 , 推測 H_a 為真

因此, 資料顯示當 income 增加 \$1000, 保險費用不會增加等量 \$1000, 而是會顯著大於收入增加的幅度